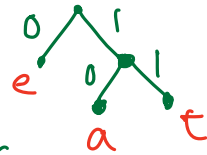
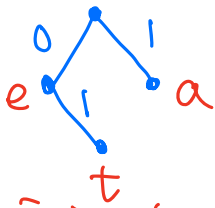


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11.2 Applications of Trees (Conti.)

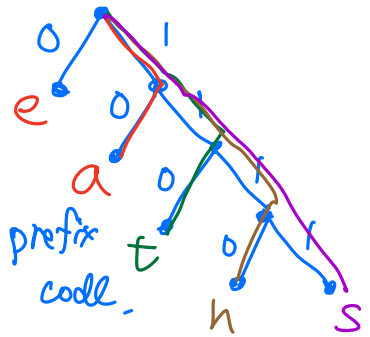
11. Build a binary tree corresponding to each given code to check if it is a prefix code:

- (1) $e: 0, a: 1, t: 01$. (2) $e: 0, a: 10, t: 11$. (3) $e: 0, a: 10, t: 110, n: 1110, s: 1111$.



Yes, it's a prefix code

It's a prefix code.



No, it is not a prefix code

Since "e" is a label of an internal vertex

12. Introduction. Game Trees

Trees can be used to analyze certain types of games such as tic-tac-toe, nim, checkers, and chess. In each of these games, two players take turns making moves. We model such games using game tree.

13. How to build a game tree?

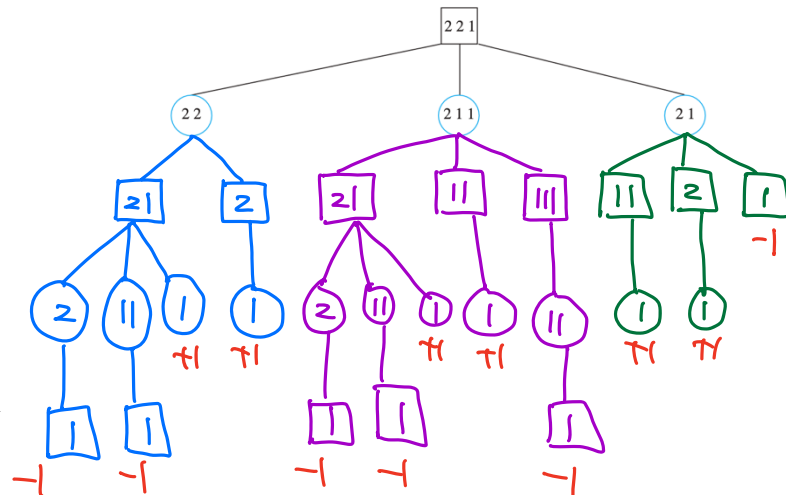
- (1) The vertices of these trees represent the positions that a game can be in as it progresses
- (2) The edges represent legal moves between these position.
- (3) The root represents the starting position.
- (4) The leaves of a game tree represent the final positions of a game.
- (5) For games that are win-lose, we label a terminal vertex represented by a circle with a "1" to indicate a win by 1st player and we label a terminal vertex represented by a box with a "-1" to indicate a win by 2nd player. For games where draws are allowed, we label a terminal vertex corresponding to a draw position with a "0".

14. Some Aspects of a Game Tree.

- (1) When the game is in a position represented by a boxed vertex at even level, it is the first player's move. When the game is in a position represented by a circle vertex at odd level, it is the second player's move.
- (2) Game trees may be *infinite* when the games they represent never end, such as games that can enter *infinite loops*, but for most games there are rules that lead to *finite game trees*.

15. Game of Nim.

- (1) Start of a game there are a number of piles of stones.
- (2) Two players take turns making moves;
- (3) A legal move consists of removing one or more stones from one of the piles, without removing all the stones left.
- (4) A player without a legal move loses (or the player removing the last stone loses)



16. Example. Complete a game tree given the starting position where there are three piles of stones containing two, two, and one stone each, respectively.

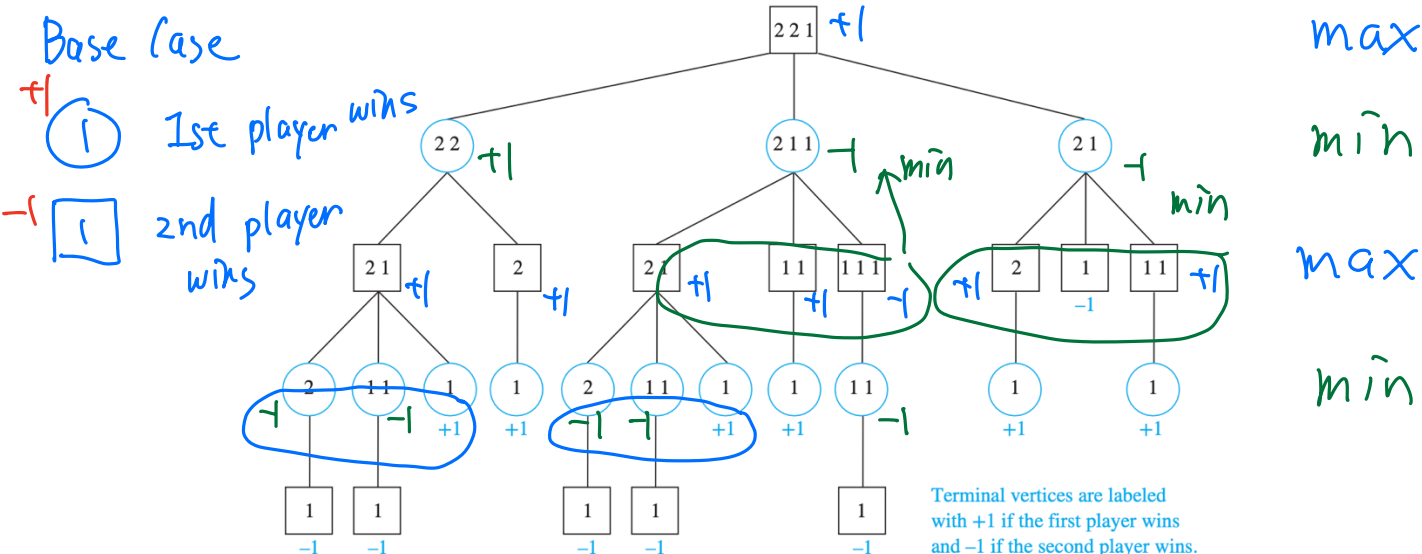
17. How to win a Game: Minimax Algorithm.

Minimax (sometimes Minmax, or saddle point) is a recursive decision-making strategy used in artificial intelligence, decision theory, combinatorial game theory, statistics, and philosophy for minimizing the possible loss for a worst case (maximum loss) scenario.

18. Steps of Minimax.

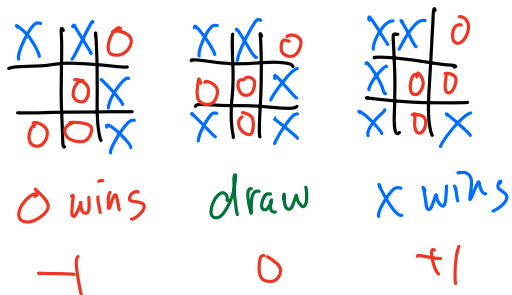
- (1) Base Case: It is the terminal state of the game and it might be win, lose, or draw. An evaluation function is used to assign to board state.
- (2) Recursive Exploration: Build a game tree with all the possible outcomes.
- (3) Backtracking: With the value of leaves assigned, we find the values of internal vertices backwardly:
 - (a) The value of an internal vertex at an even level is the maximum of the values of its children
 - (b) The value of an internal vertex at an odd level is the minimum of the values of its children

19. Use the game tree we built in Example 16 to practice the steps of Minimax.



20. Draw the subtree of the game tree for tic-tac-toe beginning at the given position. Determine the value of the subtree.

Base case:



Game tree:

